

HW # 09 Solutions

1. **Problem 1:** From HW#8, Prob. 6,

$$f^* = \sum_{i=1}^n \beta_i y_i$$

where

$$\begin{bmatrix} \langle y_1, y_1 \rangle & \langle y_2, y_1 \rangle & \cdots & \langle y_p, y_1 \rangle \\ \langle y_1, y_2 \rangle & \langle y_2, y_2 \rangle & \cdots & \langle y_p, y_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle y_1, y_p \rangle & \langle y_2, y_p \rangle & \cdots & \langle y_p, y_p \rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_p \end{bmatrix}$$

(a) We have only 1 constraint, and thus $f^* = \beta \cdot t$, where

$$\langle t, t \rangle \beta = 2$$

$$\langle t, t \rangle = \int_0^2 \tau^2 d\tau = \frac{\tau^3}{3} \Big|_0^2 = \frac{8}{3}$$

$$\therefore \frac{8}{3} \cdot \beta = 2 \Rightarrow \beta = \frac{3}{4}$$

$$\therefore \boxed{f^* = \frac{3}{4}t}$$

(b) We have 2 constraints, and thus

$$\begin{bmatrix} \langle t, t \rangle & \langle t, \sin(\pi t) \rangle \\ \langle t, \sin(\pi t) \rangle & \langle \sin(\pi t), \sin(\pi t) \rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 2 \\ \pi \end{bmatrix}$$

$$\langle t, t \rangle = \frac{8}{3} \quad \text{from part(a)}$$

$$\langle t, \sin(\pi t) \rangle = \int_0^2 \tau \sin(\pi \tau) d\tau = -\frac{2}{\pi}$$

$$\langle \sin(\pi t), \sin(\pi t) \rangle = \int_0^2 \sin^2(\pi \tau) d\tau = 1$$

$$\therefore \begin{bmatrix} \frac{8}{3} & -\frac{2}{\pi} \\ -\frac{2}{\pi} & 1 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 2 \\ \pi \end{bmatrix}$$

$$\beta_1 = \frac{3\pi^2}{2\pi^2 - 3}$$

$$\beta_2 = \frac{\pi(2\pi^2 + 3)}{2\pi^2 - 3}$$

$$f^* = \frac{3\pi^2}{2\pi^2 - 3}t + \frac{\pi(2\pi^2 + 3)}{2\pi^2 - 3}\sin(\pi t)$$

2. **Problem 2:** We define an inner product on $\mathbb{R}^n$ by $\langle x, \tilde{x} \rangle = x^\top Q \tilde{x}$, $Q > 0$. So that $\|x\| = (x^\top Q x)^{1/2}$.

Theorem If the rows of A are linearly independent, then

$$\hat{x} = \arg \min_{Ax=b} \|x\|$$

is given by

$$\hat{x} = Q^{-1} A^\top (A Q^{-1} A^\top)^{-1} b$$

Proof We decompose A into its rows, that is $A = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix}$. Define $\tilde{A} = A Q^{-1}$ and define $v_i \in \mathbb{R}^n$ by $v_i = (a_i Q^{-1})^\top = Q^{-1} a_i^\top$, where we have used the fact that Q is symmetric. It follows that

$$Ax = b \iff \langle v_i, x \rangle = b_i, \quad i = 1, \dots, p$$

Hence,

$$\hat{x} = x^* = \sum_{i=1}^p \beta_i v_i = [v_1 \mid \dots \mid v_p] \beta$$

where $G\beta = b$ and G is the Gram matrix.

Claim

$$G = A Q^{-1} A^\top \quad \text{and} \quad [v_1 \mid \dots \mid v_p] = Q^{-1} A^\top$$

□

Hence,

$$\hat{x} = Q^{-1} A^\top (A Q^{-1} A^\top)^{-1} b$$

□

Proof of Claim:

$$\begin{aligned} v_i &= Q^{-1} a_i^\top \\ [v_1 \mid \dots \mid v_p] &= [Q^{-1} a_1^\top \mid \dots \mid Q^{-1} a_p^\top] \\ &= Q^{-1} [a_1^\top \mid \dots \mid a_p^\top] \\ &= Q^{-1} A^\top \end{aligned}$$

$$\begin{aligned} G_{ij} &= \langle v_i, v_j \rangle \\ &= v_i^\top Q v_j \\ &= (a_i Q^{-1}) Q Q^{-1} a_j^\top \\ &= a_i Q^{-1} a_j^\top \end{aligned}$$

$$\begin{aligned} (A Q^{-1} A^\top)_{ij} &= a_i (Q^{-1} A^\top)_{j\text{-column}} \\ &= a_i Q^{-1} a_j^\top \end{aligned}$$

□

3. **Problem 3:** Classical Gram Schmidt applied to the columns of $A = [A_1, A_2]$ yields orthonormal $\{v^1, v^2\}$, which when placed as the columns of an orthogonal matrix yield

$$Q = \begin{bmatrix} 0.169 & 0.897 \\ 0.507 & 0.276 \\ 0.845 & -0.345 \end{bmatrix},$$

The defining $R_{ij} = \langle v^i, A_j \rangle$ completes the QR factorization developed in lecture:

$$R = \begin{bmatrix} 5.916 & 7.434 \\ 0.000 & 0.828 \end{bmatrix}.$$

Note that in general (written this way to cover all cases in this problem)

$$A_j = \sum_{i=1}^{\text{number columns of } Q} R_{ij} v^i$$

[Q,R]=qr(A) in MATLAB produces

$$Q = \begin{bmatrix} -0.169 & 0.897 & 0.408 \\ -0.507 & 0.276 & -0.817 \\ -0.845 & -0.345 & 0.408 \end{bmatrix}$$

and

$$R = \begin{bmatrix} -5.916 & -7.437 \\ 0.000 & 0.828 \\ 0.000 & 0.000 \end{bmatrix},$$

while, [Q,R]=qr(A,0), the “economy version” produces

$$Q = \begin{bmatrix} -0.169 & 0.897 \\ -0.507 & 0.276 \\ -0.845 & -0.345 \end{bmatrix},$$

and

$$R = \begin{bmatrix} -5.916 & -7.437 \\ 0.000 & 0.828 \end{bmatrix}.$$

Comparison: The economy version of the `qr` command produces an orthonormal basis for the column span of A , just as we did in class. The signs of the orthonormal vectors may be different, but otherwise, the resulting factorization is identical to what we did in lecture.

The “full” version of the `qr` command produces an orthonormal basis for the *co-domain* of A . Since A maps $\mathbb{R}^2 \rightarrow \mathbb{R}^3$, the command `[Q,R]=qr(A)` produces an orthonormal basis for $\mathbb{R}^3$; the first two columns are identical to the vectors produced by the command `[Q,R]=qr(A,0)`. The matrix R has an extra row of zeros because each column of A is now being expanded in terms of 3 vectors instead of 2. In general, if A maps $\mathbb{R}^n \rightarrow \mathbb{R}^m$, then `[Q,R]=qr(A)` produces an orthonormal basis for $\mathbb{R}^m$, and then R must be such that $QR = A$.

4. **Problem 4:** In HW2, Prob 7(b), we set $M = A^\top A$, which is symmetric. We deduce that

$$\min_{x^\top x=1} x^\top A^\top A x = \lambda_{\min}(A^\top A)$$

and

$$\arg \min_{x^\top x=1} x^\top A^\top A x = v_{\min}$$

where $v_{\min}$ is an e-vector of $A^\top A$ corresponding to the e-value $\lambda_{\min}$, i.e. $A^\top A v_{\min} = \lambda_{\min} v_{\min}$. Moreover, the length of $v_{\min}$ must be chosen to be 1 to satisfy

$$v_{\min}^\top v_{\min} = 1$$

From lecture, the columns of V are orthonormal e-vectors of $A^\top A$, arranged so that the e-values of $A^\top A$ are in decreasing order! Hence, $v_{\min}$ is the LAST column of V .

5. **Problem 6:** Let $[U, S, V] = \text{SVD}(A)$; that is

$$A = USV^\top$$

with U and V orthogonal, S is diagonal. From page 13 of the SVD handout,

$$\hat{A} = UDV^\top$$

where $D = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $S = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$

Doing the calculations in Matlab yields

$$\hat{A} = \begin{bmatrix} 4.042 & 7.045 & 3.015 \\ 10.043 & 17.034 & 7.025 \\ 16.008 & 27.003 & 11.049 \end{bmatrix}.$$

Evaluating the SVD of $\hat{A}$, per

`[U,S,V]=svd(Ahat);`

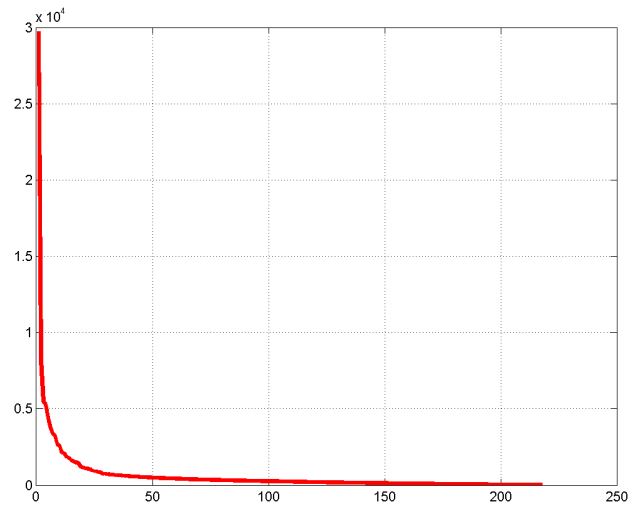
gives

$$S = \begin{bmatrix} 40.285 & 0.000 & 0.000 \\ 0.000 & 0.186 & 0.000 \\ 0.000 & 0.000 & 0.000 \end{bmatrix}$$

as expected.

6. Problem 6:

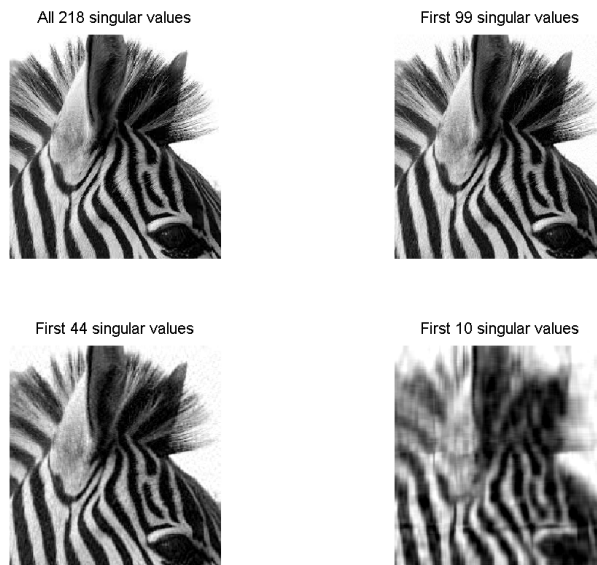
- (a) With $B = \text{double}(A)$ as in the problem statement, we do $[U, S, V] = \text{svd}(B)$. The plot of the singular values, given by $\text{diag}(s)$, shows that the singular values decrease rapidly.



Four images were generated accounting for "100%", "90%", "75%", "50%" of the area under the curve. For example

```
IntegSingVal=cumsum(diag(s));
K=IntegSingVal(end);
k2=find(IntegSingVal>.9*K);
k2=min(k2);
C2=U(:,1:k2)*S(1:k2,:)*V(1:k2,:);
```

Gives $k2=99$ and the second image. You do not need to do anything fancy! Just playing around is fine.



(b) Anything is fine.

7. Problem 7:

(a) The Jacobian is $\frac{\partial f(x)}{\partial x} = [-3x_3^3 + 6x_2, \frac{4}{3}x_2^3 + 6x_1, -9x_1x_3^2]$ and

$$\left. \frac{\partial f(x)}{\partial x} \right|_{x^*} = [21, 42, -9]$$

```
syms x1 x2 x3
x=[x1 x2 x3];
f=3*x1*(2*x2-x3^3) + (x2^4)/3;
dfdx=jacobian(f,x);
dfdx=simple(dfdx)
latex(dfdx)

x1=1; x2=3; x3=-1;
subs(dfdx)
```

(b) There is no right answer on the choice of δ . I tried $\delta = 0.01$ and obtained

```
dfdx1 = 2.1000e+01
dfdx2 = 4.2000e+01
dfdx3 = -9.0003e+00
```

I also tried $\delta = 0.05$ and obtained

```
dfdx1 = 2.1000e+01
dfdx2 = 4.2010e+01
dfdx3 = -9.0075e+00
```

which looks good too!

- (c) Hiding the function via the encryption tools in MATLAB was pretty cool, no? Iterating on the size of δ yields

$$\left. \frac{\partial f(x)}{\partial x} \right|_{x^*} = [117.3524 \quad -41.3685 \quad -636.6017 \quad -3.8520 \quad -11.2049]$$

with

$$\delta = [7.6294e-06 \quad 7.6294e-06 \quad 7.6294e-06 \quad 7.6294e-06 \quad 7.6294e-06].$$

Here is my MATLAB code. I used a WHILE loop and a FOR loop to make everything compact. Note how the standard basis vectors were generated as rows of the identity matrix (would have done columns if I had made my x_0 a column vector).

```
err_max=1;
E=eye(5);
dfdx=zeros(1,5);
x0=[1 1 1 1 1];
delta=1*[1 1 1 1 1];
while err_max>1e-4
    delta=delta/2;
    dfdx_prev=dfdx;
    for k=1:5
        dfdx(k)=( funcPartC(x0+delta(k)*E(k,:)) - ...
            funcPartC(x0-delta(k)*E(k,:)) )/2/delta(k);
    end
    error = dfdx-dfdx_prev
    err_max=max(abs(error));
end

delta
dfdx
```

8. Problem 8:

Prediction Step:

$$\hat{x}_{1|0} = \hat{x}_{0|0} + \delta_t \hat{u}_1 = 1 + 0.1 \times 10 = 2$$

$$P_{1|0} = P_{0|0} + \delta_t \times R \times \delta_t = 0.25 + 0.1 \times 16 \times 0.1 = 0.41$$

Measurement Update Step:

$$K_1 = P_{1|0} C^\top (C P_{1|0} C^\top + Q)^{-1} = 0.41 \times \frac{-2}{c} \times \left(\frac{-2}{c} \times 0.41 \times \frac{-2}{c} + 10^{-18} \right)^{-1} = -1.422 \times 10^8$$

$$\hat{z}_{1|0} = \frac{2}{c} (5 - \hat{x}_{1|0}) = \frac{2}{3 \times 10^8} (5 - 2) = 2 \times 10^{-8}$$

$$\hat{x}_{1|1} = \hat{x}_{1|0} + K_1 (z_1 - \hat{z}_{1|0}) = 2 - 1.422 \times 10^8 \times (2.2 \times 10^{-8} - 2 \times 10^{-8}) = 1.7156$$

$$P_{1|1} = P_{1|0} - K_1 C P_{1|0} = 0.41 - (-1.422) \times 10^8 * \frac{-2}{3 \times 10^8} = \times 0.41 = 0.0213$$

Assembling everything together, the state x_1 can now be estimated as $x_1 \sim \mathcal{N}(1.7156, 0.0213)$.

Note 1: In this problem, we trust the sensor measurement a lot and hence you can see that the covariance has decreased a lot. You can try $Q = 10^{-17}$ and see how the state estimate, x_1 changes.

Note 2: $P_{1|1}$ is always less than $P_{1|0}$, otherwise, you would not be using the filter!